

COURSE NOTES

Combinatorics II

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2026 年 3 月 3 日

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第一章 Lecture 1

Basic Information

Office hours: Mon, 13:45-15:45.

Main book: *A course in combinatorics* by J.H. van Lint and R.M. Wilson, 2nd edition. And the lecture notes by Prof. Ihringer.

Grade: Final Exam: 40%, midterm 30 % (April 15, week 8 in class), Homework 20%, Quizzes 10% (very easy based on HW).

Homework:

- Biweekly
- Only handwritten on paper
- Submission in class
- Only in English

Preliminary Schedule:

- **Week 1-4:** Basic graph theory, including trees, Ramsey numbers, Turan numbers, Hall's marriage theorem, etc.
- **Week 5-6:** Hadamard matrices.
- **Week 6-8:** Projective spaces.
- **Week 9:** Designs.
- **Week 10:** Strongly regular graphs.
- **Week 11:** Generalized Quadrangles.
- **Week 12-13:** General Spectral Graph Theory.
- **Week 14:** Associated Schemes.
- **Week 15:** Coding Theory.

1.1 Introduction to Graphs

Definition 1.1.1. A graph G is a pair (V, E) where V is a set of vertices and E is a set of edges.

Let $G = (V, E)$ be a graph with vertex set V and edge set E .

- If G is directed, then edges are ordered pairs and $E \subseteq V \times V$;
- If G is simple undirected, then edges are unordered pairs and $E \subseteq \binom{V}{2}$. (We call such G a normal graph in our course.)

Definition 1.1.2. Let $G = (V, E)$ be a graph.

- Let $e = \{x, y\} \in E$. The vertices x and y are said to be **incident** with the edge e . Two vertices x, y are called **adjacent** if $\{x, y\} \in E$.
- A vertex is called an **isolated vertex** if it is not incident with any edge.
- An edge connecting a vertex to itself is called a **loop**.
- Two or more edges connecting the same pair of vertices are called **multiple edges**.

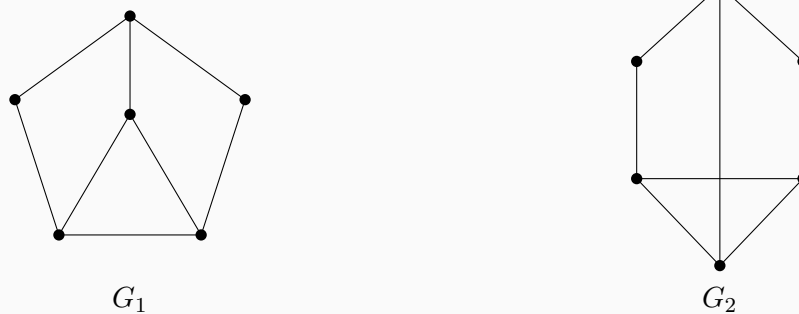
Example 1.1.3. The following figure illustrates the definitions:



- **Incident/Adjacent:** Vertices x and y are adjacent and incident to edge e .
- **Isolated:** Vertex z has no incident edges.
- **Loop:** Vertex u has an edge connecting to itself.
- **Multiple Edges:** Vertices v and w are connected by two edges.

Definition 1.1.4. A graph is called a **planar graph** if it possesses a planar drawing, that is, it can be drawn in the plane such that no two edges intersect except at their endpoints.

Example 1.1.5. Consider the following two drawings:



G_1 and G_2 are **isomorphic**: If we pull the center node of G_1 then we get G_2 . G_1 is planar, but G_2 is not. We call G_1 the planar drawing of G_2 , which means that G_1 has no edges intersect.

Definition 1.1.6. A graph without loops and multiple edges is called a **simple graph**.

Remark 1.1.7. In the two statements above, the intended notion is essentially the same. The condition “ G is simple undirected” together with $E \subseteq \binom{V}{2}$ means that every edge is an unordered pair $\{u, v\}$ of distinct vertices, so loops are excluded automatically, and since E is a set (rather than a multiset), multiple edges are excluded as well. Hence this coincides with the standard definition: a simple graph is a graph with no loops and no multiple edges.

In our course notes, such graphs are called normal graphs. The only minor caveat is that some texts use “simple graph” with the convention that graphs are undirected by default, whereas others may state “simple undirected” explicitly. In the present context, these conventions agree.

Convention: All our graphs are simple. Unless we say otherwise.

Definition 1.1.8 (Map coloring and the associated graph). A map (on the plane or on the sphere) is a decomposition of the surface into finitely many regions (“countries”) such that any two regions meet either in a boundary arc, in finitely many boundary points, or not at all. Two countries are said to be adjacent if they share a boundary segment of positive length (meeting at a single point does not count). Given such a map, we define its adjacency graph (also called the dual graph) $G = (V, E)$ as follows:

$$V = \{\text{countries of the map}\}, \quad \{u, v\} \in E \iff \text{countries } u \text{ and } v \text{ are adjacent.}$$

Remark 1.1.9 (Planarity). The adjacency graph G is planar: it can be drawn in the plane with no edge crossings by placing a vertex inside each country and drawing an edge between two vertices through the common boundary segment of the corresponding countries. Thus the problem of coloring maps is equivalent to coloring planar graphs.

Definition 1.1.10 (Vertex coloring and chromatic number). Let $G = (V, E)$ be a graph. A (proper) vertex coloring of G with k colors is a map $c : V \rightarrow \{1, 2, \dots, k\}$ such that

$$\{u, v\} \in E \implies c(u) \neq c(v).$$

The smallest k for which such a coloring exists is called the chromatic number of G and is denoted by $\chi(G)$.

Theorem 1.1.11 (Four-Color Theorem). For every planar graph G , one has

$$\chi(G) \leq 4.$$

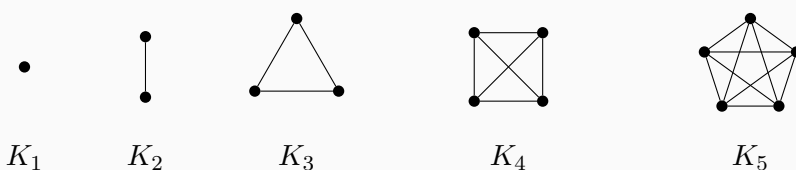
Equivalently, every planar map can be colored with at most four colors so that any two adjacent countries receive different colors.

Remark 1.1.12 (Why “at most 5 colors” is easy). *A classical result (the Five-Color Theorem) states that every planar graph satisfies $\chi(G) \leq 5$. This theorem admits a relatively short proof using standard planar graph tools such as Euler’s formula and a reducible configuration argument (often presented via induction on $|V|$). By contrast, the Four-Color Theorem is substantially deeper and historically required extensive case-checking (computer-assisted) in its first proofs.*

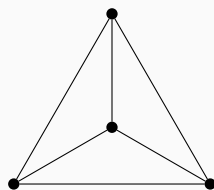
The following statements are equivalent:

$$\{x, y\} \in E \Leftrightarrow x \sim y \Leftrightarrow x, y \text{ adjacent} \Leftrightarrow x, y \text{ are neighbors}$$

Example 1.1.13. $|V| = n$, $E = \binom{V}{2}$. Complete graph K_n .

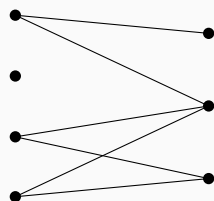


Remark: K_4 is a planar graph, but K_5 is not.

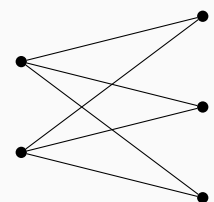


Planar drawing of K_4

Example 1.1.14. $G = (V_1 \cup V_2, E)$ is bipartite if $E \cap (\binom{V_1}{2} \cup \binom{V_2}{2}) = \emptyset$.

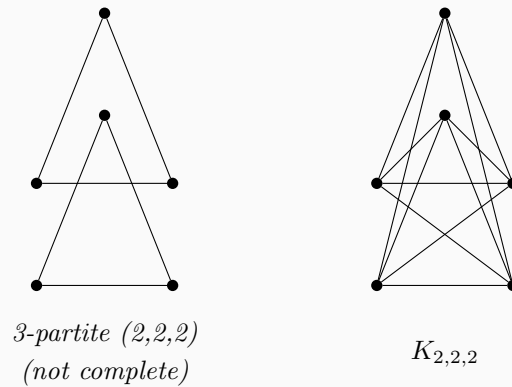


Example 1.1.15. If $E = \{\{x, y\} : x \in V_1, y \in V_2\}$, then G is a complete bipartite graph, denoted by $K_{m,n}$ where $m = |V_1|, n = |V_2|$.

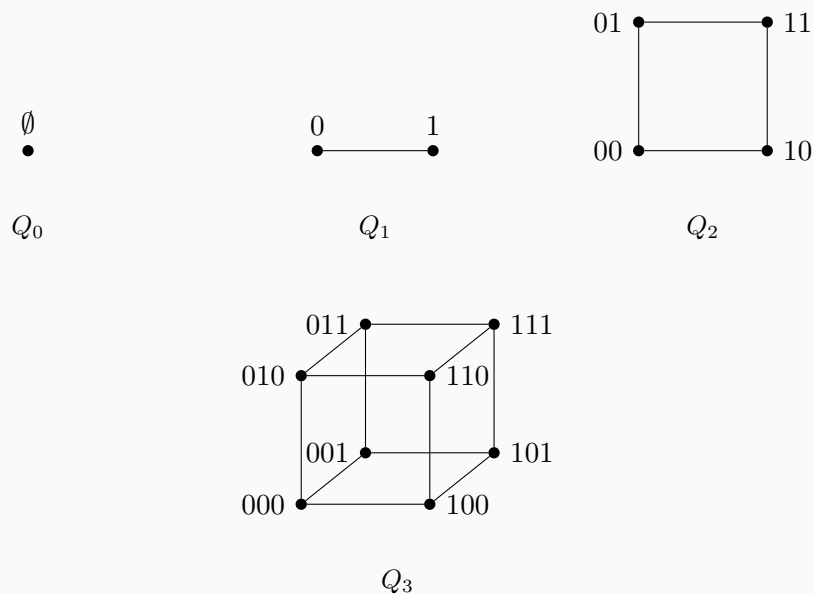


$K_{2,3}$

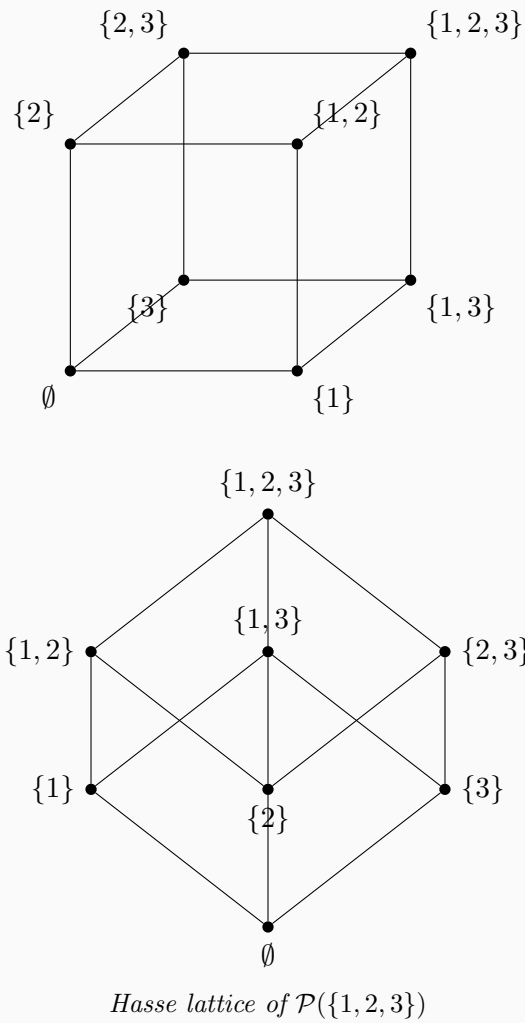
Example 1.1.16. Similarly we can define k -partite graphs and complete k -partite graphs.



Example 1.1.17. The n -dimensional hypercube Q_n is the graph with vertex set $V(Q_n) = \{0,1\}^n$, the set of binary strings of length n . Two vertices $x, y \in V(Q_n)$ are adjacent if and only if they differ in exactly one coordinate (i.e., their Hamming distance is 1). The graph Q_n is n -regular with $|V(Q_n)| = 2^n$ and $|E(Q_n)| = n2^{n-1}$. For $n = 3$, we have $110 \sim 111$, whereas $111 \not\sim 100$.

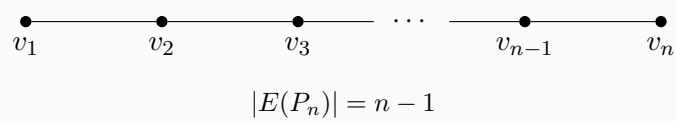


Example 1.1.18. Let S be a set with 3 elements, for instance $S = \{1,2,3\}$. The power set $\mathcal{P}(S)$ can be made into a graph by defining two subsets $A, B \subseteq S$ to be adjacent if their symmetric difference has size one, i.e., $|A \Delta B| = 1$. This graph is isomorphic to the 3-dimensional hypercube Q_3 . The structure is often drawn as the Hasse diagram of the subset lattice.

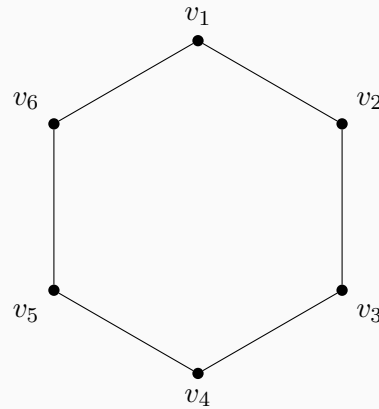


This graph is another representation of Q_3 .

Example 1.1.19. P_n : path of length $n - 1$.

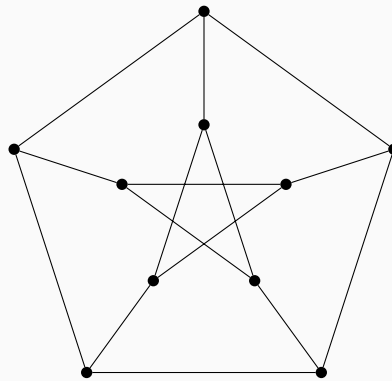


Example 1.1.20. With $\{v_n, v_1\}$ added, we get a cycle C_n . For example C_6 :



$$|V(C_6)| = |E(C_6)| = 6$$

Example 1.1.21. *Petersen Graph:*



Petersen graph

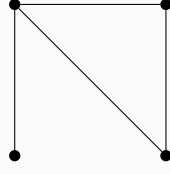
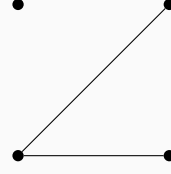
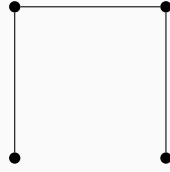
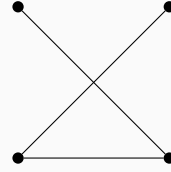
Each vertex has degree 3 so this graph is 3-regular. Any two vertices linked by an edge has no common neighbor. Any 2 vertices not linked by an edge has a common neighbor.

It is a strongly regular graph with parameters $(10, 3, 0, 1)$.

Definition 1.1.22 (Strongly regular graph). Let $G = (V, E)$ be a simple undirected graph. We say that G is strongly regular with parameters (v, k, λ, μ) if the following hold:

1. $|V| = v$ and G is k -regular (i.e. every vertex has degree k);
2. for any two adjacent vertices $x \sim y$, the number of common neighbors of x and y is exactly λ , i.e. $|\Gamma(x) \cap \Gamma(y)| = \lambda$;
3. for any two non-adjacent vertices $x \not\sim y$, the number of common neighbors of x and y is exactly μ , i.e. $|\Gamma(x) \cap \Gamma(y)| = \mu$.

Example 1.1.23. $G = (V, E)$, $\bar{G} = (V, \bar{E})$, where $\bar{E} = \binom{V}{2} \setminus E$. $\bar{G}$ is the complement of G .

 G  $\bar{G}$  P_4  $\bar{P}_4$

In this example, observe that $P_4 \cong \bar{P}_4$. We call such a graph self-complementary. A graph G is self-complementary if $G \cong \bar{G}$.

Definition 1.1.24. Let $G = (V, E)$ and $G' = (V', E')$. They are isomorphic if and only if there exists a bijection $\phi : V \rightarrow V'$ such that $\{x, y\} \in E \Leftrightarrow \{\phi(x), \phi(y)\} \in E'$.

Definition 1.1.25. $\deg(x) : x \sim y = \text{degree of } x$, denoted by $\deg(x)$ or $d(x)$. If $d(x) = d(y)$ for all $x, y \in V$, then G is d -regular.

Remark 1.1.26 (Degree sequence: definition and uses). Let $G = (V, E)$ be a simple undirected graph with $n = |V|$. The degree sequence of G is the list of vertex degrees arranged in nonincreasing order,

$$d_1 \geq d_2 \geq \cdots \geq d_n, \quad \text{where } \{d_1, \dots, d_n\} = \{\deg(v) : v \in V\}.$$

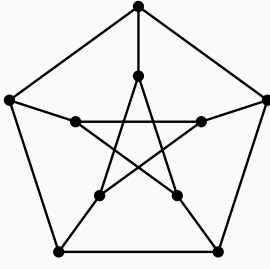
It is useful for:

- **Coarse structural information:** by the Handshaking Lemma, $\sum_{i=1}^n d_i = 2|E|$, so the degree sequence determines the number of edges and reveals properties such as regularity (all d_i equal).
- **Realizability tests:** deciding whether a given integer sequence is graphical (i.e. occurs as the degree sequence of a simple graph), using criteria such as the Erdős–Gallai theorem or the Havel–Hakimi algorithm (which can also construct a realizing graph).

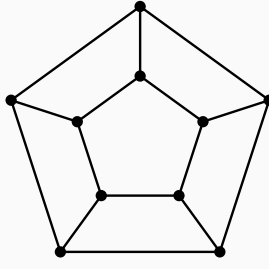
- **Isomorphism invariant:** isomorphic graphs have the same degree sequence, so differing degree sequences imply non-isomorphism, though the converse need not hold.
- **Algorithmic and modeling input:** degree sequences serve as compact summaries in extremal graph theory, network models (e.g. random graphs with prescribed degrees), and matrix-based methods (e.g. Laplacian $L = D - A$ where D is the degree matrix).

Example 1.1.27. $d_1 = 2, d_2 = 1, d_3 = 1$. This graph is P_3 .

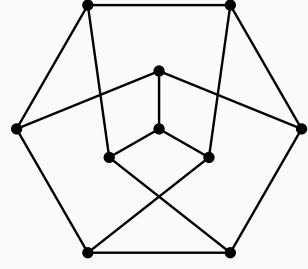
Example 1.1.28. Consider the following 3 graphs:



G_1



G_2



G_3

$G_1 \not\cong G_2$, $G_2 \not\cong G_3$, but $G_1 \cong G_3$.

The *graph isomorphism problem* (GI) asks: given two finite graphs G and H , decide whether there exists a bijection between their vertex sets that preserves adjacency (i.e. an isomorphism). From the viewpoint of worst-case complexity, GI has long occupied an intermediate position: it is in NP, but it is neither known to be NP-complete nor known to admit a polynomial-time algorithm. A major milestone is Babai's 2015 breakthrough, which gives a *quasi-polynomial* time algorithm for GI, running in time $\exp((\log n)^{O(1)})$ on n -vertex graphs, substantially improving the best previously known general upper bounds.

In parallel with the worst-case theory, there is a large body of *practical* work aimed at solving typical instances efficiently, often via *canonical labeling* and computation of automorphism groups. Since the 1990s, McKay's software **nauty** (and later **Traces**, with further algorithmic refinements) has been highly effective on a wide range of graphs encountered in applications. However, strong empirical performance does not settle the worst-case complexity question, and there exist carefully constructed families of graphs that can be difficult for certain refinement-and-search heuristics. Thus, it is important to distinguish clearly between (i) the unresolved *theoretical* question of whether GI is solvable in polynomial time in the worst case, and (ii) the largely successful but still evolving *algorithm engineering* of fast GI solvers for real-world inputs.

Proposition 1.1.29 (handshaking lemma). $G = (V, E)$. Then

$$\sum_{v \in V} \deg(v) = 2|E|$$

证明. Count $(x, e) \in V \times E$ where $x \in e$ in two ways. First, for each $e \in E$, there are exactly two vertices $x \in e$. So the number of such pairs is $2|E|$. Second, for each $x \in V$, there are exactly $\deg(x)$ edges $e \in E$ containing x . So the number of such pairs is $\sum_{v \in V} \deg(v)$. Equating these two counts gives the desired result. $\square$

Theorem 1.1.30. A finite graph (we only talk about finite graphs in our class) has an even number of vertices with odd degree.

证明. Consider

$$\sum_{v \in V} \deg(v) = 2|E|,$$

which is an even integer.

Reduce this congruence modulo 2. Vertices of even degree contribute 0 modulo 2, and vertices of odd degree contribute 1 modulo 2. Hence

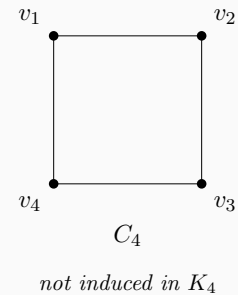
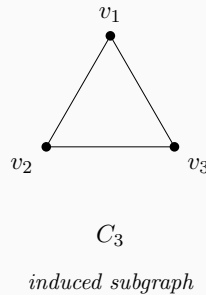
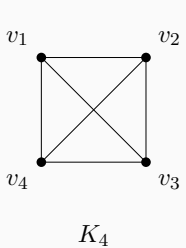
$$\sum_{v \in V} \deg(v) \equiv \#\{v \in V : \deg(v) \text{ is odd}\} \pmod{2}.$$

Since the left-hand side is even, the right-hand side is 0 (mod 2), so the number of odd-degree vertices is even. $\square$

Definition 1.1.31 (subgraph). Let $G = (V, E)$ be a graph. A graph $H = (V', E')$ is a subgraph of G if $V' \subseteq V$ and $E' \subseteq E$.

Definition 1.1.32 (induced subgraph). Let $G = (V, E)$ be a graph. A graph $H = (V', E')$ is an induced subgraph of G if $V' \subseteq V$ and $E' = E \cap \binom{V'}{2}$.

Example 1.1.33. Consider the graph K_4 , we have two subgraph C_3 and C_4 . The subgraph C_3 is an induced subgraph, but the subgraph C_4 is not.



For C_3 , we choose the vertices $\{v_1, v_2, v_3\}$, and every edge among them in K_4 is kept,

so it is induced. For C_4 , we use all four vertices but delete the two diagonals, so it is a subgraph but not an induced subgraph.

Adjacency list representation.

Let $G = (V, E)$ be a (finite) graph with $V = \{1, 2, \dots, n\}$. In an *adjacency list* representation, the computer stores an array (or map) Adj indexed by vertices, where $\text{Adj}[u]$ is the list of all neighbors of u :

$$\text{Adj}[u] = \{v \in V : \{u, v\} \in E\} \quad (\text{undirected}), \text{Adj}[u] = \{v \in V : (u, v) \in E\} \quad (\text{directed}).$$

For an undirected simple graph each edge $\{u, v\}$ appears twice, once in $\text{Adj}[u]$ and once in $\text{Adj}[v]$. This data structure supports efficient traversal of neighbors (e.g. BFS/DFS) in total time $O(|V| + |E|)$ and uses space $O(|V| + |E|)$ for sparse graphs.

For our human beings we use the following matrices:

Definition 1.1.34 (adjacent matrix). Let $G = (V, E)$ be a graph. Let $V = \{v_1, v_2, \dots, v_n\}$ and $E = \{e_1, e_2, \dots, e_m\}$. Let $A = (a_{ij})$ where $a_{ij} = 1$ if $v_i \sim v_j$, and $a_{ij} = 0$ otherwise. Then A is called the adjacent matrix of G . For the base field usually we use $\mathbb{R}, \mathbb{C}, \mathbb{F}_2, \mathbb{F}_q$.

Remark 1.1.35. In our class we mainly use consider $A \in \mathbb{R}^{n \times n}$. Sometimes we will consider $A \in \mathbb{C}^{n \times n}$ when we move to spectral graph theory.

Definition 1.1.36 (Incidence matrix). Let $B = (b_{ij})$, $B \in \mathbb{R}^{n \times m}$. $b_{ij} = 1$ if vertex v_i is incident to edge e_j , and $b_{ij} = 0$ otherwise. Then B is called the incidence matrix of G .

Definition 1.1.37 (degree matrix). Let $D = (d_{ij})$ be a diagonal matrix where $d_{ii} = \deg(v_i)$, and $d_{ij} = 0$ if $i \neq j$. Then D is called the degree matrix of G .

Exercise 1.1.38. Prove that $A = BB^T - D$.

证明. We assume throughout that $G = (V, E)$ is a finite *simple* undirected graph (no loops, no multiple edges) and that B is the (non-oriented) incidence matrix defined by

$$b_{ij} = \begin{cases} 1, & \text{if } v_i \text{ is incident to } e_j, \\ 0, & \text{otherwise.} \end{cases}$$

Let $M := BB^T = (m_{ik})$. Then

$$m_{ik} = (BB^T)_{ik} = \sum_{j=1}^m b_{ij} b_{kj}.$$

For fixed vertices v_i, v_k , the product $b_{ij}b_{kj}$ equals 1 precisely when *both* v_i and v_k are incident

to the same edge e_j , and equals 0 otherwise. Hence m_{ik} counts the number of edges that are incident to both v_i and v_k .

Case 1: $i = k$. Then m_{ii} counts the number of edges incident to v_i , so

$$m_{ii} = \deg(v_i) = d_{ii}.$$

Case 2: $i \neq k$. In a simple graph, there is *at most one* edge joining v_i and v_k . Therefore

$$m_{ik} = \begin{cases} 1, & \text{if } v_i \sim v_k, \\ 0, & \text{otherwise,} \end{cases}$$

so for $i \neq k$ we have $m_{ik} = a_{ik}$.

Combining the two cases, we see that BB^T agrees with A on all off-diagonal entries, and has diagonal entries $\deg(v_i)$. Equivalently,

$$BB^T = A + D,$$

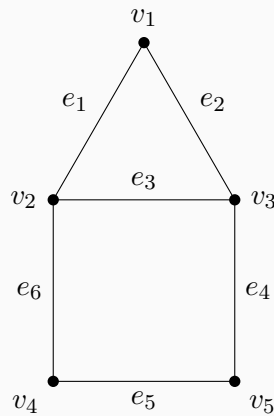
and therefore

$$A = BB^T - D,$$

as required. □

Definition 1.1.39 (Laplacian matrix). *Let $L = D - A$, where D is the degree matrix and A is the adjacency matrix. Then L is called the Laplacian matrix of G .*

Example 1.1.40. *Consider this graph:*



With the vertex order $(v_1, v_2, v_3, v_4, v_5)$ and the edge order $(e_1, e_2, e_3, e_4, e_5, e_6)$, the

corresponding matrices are:

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

$$B = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

$$D = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

$$L = D - A = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 \\ -1 & -1 & 3 & 0 & -1 \\ 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

We can also verify the exercise $A = BB^T - D$ directly:

$$BB^T = \begin{pmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 3 & 1 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix}.$$

Hence

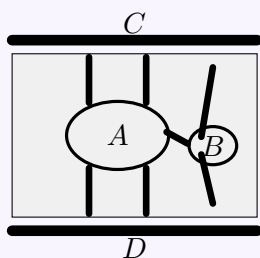
$$BB^T - D = \begin{pmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 3 & 1 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix} = A.$$

Definition 1.1.41 (*k*-walk). Let $G = (V, E)$ be a graph. A *k*-walk in G is a sequence of vertices $(v_0, v_1, \dots, v_k)$ such that $\{v_i, v_{i+1}\} \in E$ for all $i = 0, 1, \dots, k-1$. A walk

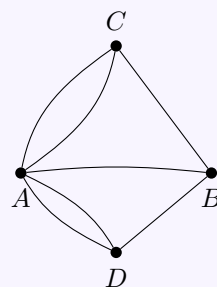
is closed if $v_0 = v_k$. If a k -walk is a path, then all vertices are distinct. If a k -walk is a cycle, then all vertices are distinct except for the first and last vertices.

第二章 Lecture 2

Topic 2.0.1 (The Seven Bridges of Königsberg). *In Königsberg (now Kaliningrad), the river Pregel passed through the city with two islands and seven bridges connecting the islands and the two banks (see figure). The problem: can one start at some place, cross each bridge exactly once, and return to the start? Euler (1736) abstracted the two banks and two islands as four vertices and the seven bridges as seven edges, obtaining the graph G ; he proved that such a closed trail exists if and only if every vertex has even degree. In G all four vertices have odd degree, so no solution exists. This problem is regarded as the origin of graph theory and Eulerian trails.*



Schematic of the seven bridges



Abstract graph G

Definition 2.0.2. *An Eulerian path is a walk which contains each edge of a graph exactly once.*

Definition 2.0.3. *An Eulerian cycle is a closed Eulerian path. A graph is called Eulerian if it contains an Eulerian cycle.*

Lemma 2.0.4. *Let $G = (V, E)$ be a graph with $|E| \geq 1$. If every vertex has degree at least 2 (i.e., $\deg(v) \geq 2$ for all $v \in V$), then G contains a cycle.*

证明. Take a maximal path $P = (v_0, v_1, \dots, v_k)$ in G (so no edge extends P at either end). Since $|E| \geq 1$, we have $k \geq 1$. The vertex v_k has $\deg(v_k) \geq 2$, so v_k has a neighbor other than v_{k-1} . Any such neighbor must lie on P : if v_k were adjacent to some $u \notin \{v_0, \dots, v_{k-1}\}$, then P could be extended to $v_0 \cdots v_k u$, contradicting maximality. So v_k is adjacent to v_i for some i with $0 \leq i \leq k-2$. Then $(v_i, v_{i+1}, \dots, v_k, v_i)$ is a cycle in G . $\square$

Theorem 2.0.5. *A finite connected graph is Eulerian if and only if every vertex has even degree.*

证明. *Necessity* ($\Rightarrow$). Let $C = (x_0, x_1, \dots, x_n)$ be an Eulerian cycle with $x_0 = x_n$. Each edge is used exactly once. For any vertex $y \in V$, each time C passes through y it uses one edge to enter and one to leave; so the number of edges incident to y used by C is twice the number of times y appears in $(x_1, \dots, x_{n-1})$. The start/end vertex $x_0 = x_n$ is left once and entered once, contributing 2 more. Thus $\deg(y)$ is even for all y .

Sufficiency ($\Leftarrow$). (Hierholzer, 1873.) We use induction on $|E|$. If $|E| = 0$, then G is a single vertex and the trivial closed walk is an Eulerian cycle. Suppose $|E| \geq 1$. Since G is connected and every vertex has even degree, $\delta(G) \geq 2$, so G contains a cycle by the lemma above. Start at any vertex x and take a maximal walk that does not repeat any edge. Whenever the walk enters a vertex other than x , it has used an odd number of edges at that vertex so far, so an odd number (hence at least one) remains; thus the walk cannot get stuck except at x . So the walk returns to x , forming a closed trail C_1 with edge set $E_1 \subseteq E$.

Remove E_1 from G . In $G' = (V, E \setminus E_1)$, every vertex still has even degree (each vertex on C_1 lost 0 or 2 incident edges). So every non-trivial component of G' has an Eulerian cycle by the induction hypothesis. If $E_1 = E$, then C_1 is an Eulerian cycle. Otherwise, pick a vertex v that lies on C_1 and belongs to a component H of G' with at least one edge. That component has an Eulerian cycle C_2 (starting and ending at v). Splice C_2 into C_1 at v : traverse C_1 until v , then traverse all of C_2 , then continue C_1 from v . Repeating this for every such component yields an Eulerian cycle of G . The algorithm runs in $O(|E|)$ time. $\square$

Example 2.0.6. *Figure 3*

Definition 2.0.7. *A Hamiltonian cycle is a closed walk which uses every vertex precisely once. A graph is called Hamiltonian if it contains a Hamiltonian cycle.*

If all degrees $\geq \frac{n+1}{2}$, then G is Hamiltonian.

Random graphs are Hamiltonian.

In general, problem of NP-complete.

P: det-turing machine can solve in polynomial time. NP: non-deterministic turing machine can solve in polynomial time.

NP-complete: any problem in NP can be reduced to an NP-complete in polynomial time.

Graph theory: structure graph theory, algorithmic graph theory, extremal graph theory...

Now we talk about trees.

Definition 2.0.8. *A connected graph with no cycles is called a tree.*

Figure 4

Definition 2.0.9. *Disjoint union of trees is called forest.*

Figure 5

Definition 2.0.10. *For a connected graph $G = (V, E)$, a subgraph $T = (V, E')$ a tree is called the spanning tree of G .*

The spanning tree of G always exists.

Theorem 2.0.11. *The followings are equivalent:*

1. $G = (V, E)$ is a tree.
2. Any two vertices are connected by a unique path.
3. G is connected with $|V| - 1 = |E|$.

证明. ((1) $\Rightarrow$ (2)): If x, y are connected by two distinct paths, then we find a cycle.

((2) $\Rightarrow$ (1)): If G contains a cycle, then two vertices on the cycle are connected by two distinct paths. (The connectivity is clear).

((1) $\Rightarrow$ (3)): $|V| = 1$, true. Now $|V| > 1$, let $P = (u, u_1, \dots, u_l)$ be a longest path in G , then all neighbors of u are in P and u must be a leaf. Remove the leaf, by induction we prove it.

((3) $\Rightarrow$ (1)): G is connected so G has a spanning tree T , then $1 = |V(G)| - |E(T)| \geq |V(G)| - |E(G)| = 1$, hence $E(G) = E(T)$ which implies $G = T$ is a tree. $\square$

Corollary 2.0.12. *G forest, # connected components of $G = |V(G)| - |E(G)|$.*

Corollary 2.0.13. *Let T be a tree, $n := |V(T)| \geq 2$ with degree sequence $(d_1, \dots, d_n)$.
Figure 5*

What is the number of trees on n vertices? $t(n)$. Cayley's formula.

Figure 6

$t(4) = 16$

Theorem 2.0.14. $t(n) = n^{n-2}$.

证明. Goal: find a bijection between trees on sets $[n]$ and $[n]^{n-2}$.

Let T be a tree

Figure 7

1. $u_1 \in u = \min\{v \in V : \deg(v) = 1\}$, $a_1 :=$ unique neighbor of u_1 .
2. Remove u_1 and repeat.

We get a sequence $a = (a_1, \dots, a_{n-2})$ in $[n]^{n-2}$.

To show: For each sequence a in $[n]^{n-2}$, there is a unique tree T .

Let $d_i := \#$ degree of vertex i and $f_i := \#$ of occurrences of i in a .

Note that $f_i \leq d_i - 1$ for all i . Hence $n - 2 = \sum f_i \leq \sum (d_i - 1) = (2n - 2) - n = n - 2$.

Hence $f_i = d_i - 1$ for all i .

We construct T :

1. Smallest b_1 not in $(a_1, \dots, a_{n-2})$ and add edge (a_1, b_1)
2. Take $b_2 \neq b_1$ and not in $(a_2, \dots, a_{n-2})$ and add edge (a_2, b_2) . Continue this process any finally we construct a tree.

□

证明. $t(n : d_1, \dots, d_n) = \#$ trees with degrees $d_1, \dots, d_n$.

Recall $\binom{n}{r_1 \dots r_k} = \sum_{i=1}^{n-1} \binom{n-1}{r_1 \dots r_{i-1} r_{i+1}} \dots r_k$.

Then $t(n : d_1, \dots, d_n) = \sum_{i=1}^n t(n-1 : d_1, \dots, d_{i-1}, d_{i+1}, \dots, d_n)$. WLOG, $d_n = 1$ with basis $t(3 : d_1, d_2, d_3) = \binom{1}{d_1-1 d_2-1 d_3-1}$. We can prove $t(n : d_1, \dots, d_n) = \binom{n-2}{d_1-1 \dots d_n-1}$. Using $(x_1 + \dots + x_k)^n = \sum \binom{n}{r_1 \dots r_k} x_1^{r_1} \dots x_k^{r_k}$, we get $t(n) = n^{n-2}$ □

证明. Kirchoff's matrix tree theorem: $\#$ spanning trees of G is equal to $\frac{1}{n} \cdot \lambda_1 \dots \lambda_{n-1}$. Where $\lambda_1, \dots, \lambda_{n-1}$ non-zero eigenvalues of $L(G)$.

$t(n) = \#$ spanning trees of K_n . $L(K_n) = nI - J$. Eigenvalues $n-1$ times n , 1 times 0. Hence $t(n) = \frac{1}{n}(n-1)^{n-1} = n^{n-2}$. □

see more proofs on your book.

第三章 Homework

3.1 Week 1

Exercise 3.1.1. Let A be the adjacency matrix of a graph $G = (V, E)$.

(a) Find a combinatorial interpretation for each entry of A^2 and A^3 .

(b) Show that $\text{tr}(A^2) = 2|E|$.

(c) For $m \geq 2$ an integer, what does $\text{tr}(A^m)$ count? Prove your claim.

证明. (a) For $k \geq 1$ we have

$$(A^k)_{ij} = \sum_{t_1, \dots, t_{k-1}} a_{it_1} a_{t_1 t_2} \cdots a_{t_{k-1} j}.$$

Each product $a_{it_1} a_{t_1 t_2} \cdots a_{t_{k-1} j}$ equals 1 precisely when

$$v_i \sim v_{t_1} \sim v_{t_2} \sim \cdots \sim v_{t_{k-1}} \sim v_j,$$

i.e. when $(v_i, v_{t_1}, \dots, v_{t_{k-1}}, v_j)$ is a walk of length k from v_i to v_j ; otherwise the product is 0. Hence $(A^k)_{ij}$ counts the number of walks of length k from v_i to v_j .

In particular:

$$(A^2)_{ij} = \sum_{t=1}^n a_{it} a_{tj}$$

counts the number of vertices v_t adjacent to both v_i and v_j , i.e. the number of length-2 walks $v_i \rightarrow v_t \rightarrow v_j$.

Similarly,

$$(A^3)_{ij} = \sum_{s,t=1}^n a_{is} a_{st} a_{tj}$$

counts the number of length-3 walks $v_i \rightarrow v_s \rightarrow v_t \rightarrow v_j$.

(b) Since $a_{ii} = 0$ for a simple graph, we have

$$(A^2)_{ii} = \sum_{t=1}^n a_{it} a_{ti} = \sum_{t=1}^n a_{it}^2 = \sum_{t=1}^n a_{it} = \deg(v_i).$$

Therefore

$$\text{tr}(A^2) = \sum_{i=1}^n (A^2)_{ii} = \sum_{i=1}^n \deg(v_i).$$

By the Handshaking Lemma, $\sum_{i=1}^n \deg(v_i) = 2|E|$. Hence

$$\text{tr}(A^2) = 2|E| \quad (\text{equivalently, } \text{tr}(A^2) - 2|E| = 0).$$

(c) For any integer $m \geq 1$, by the argument in (a),

$(A^m)_{ij}$ counts the number of walks of length m from v_i to v_j .

In particular, $(A^m)_{ii}$ counts the number of *closed* walks of length m starting and ending at v_i . Summing over i yields

$$\text{tr}(A^m) = \sum_{i=1}^n (A^m)_{ii},$$

so $\text{tr}(A^m)$ counts the total number of closed walks of length m in G , where a closed walk is counted with its chosen starting vertex (i.e. each closed walk contributes 1 to exactly one diagonal entry, the entry corresponding to its start/end vertex).

□

Exercise 3.1.2. *The girth of a graph G is the length of the smallest polygon in the graph. Let G be a graph of girth 5 for which all vertices have degree at least d . Show that G has at least $d^2 + 1$ vertices. Can equality hold?*

证明. Fix a vertex $x \in V(G)$ and write $N(x) = \{y_1, \dots, y_t\}$ as the set of neighbors of x , where $t = \deg(x) \geq d$.

Step 1: no triangles. Since the girth is 5, G has no 3-cycles. Hence the neighbors of x are pairwise nonadjacent: $y_i \not\sim y_j$ for $i \neq j$.

Step 2: no 4-cycles. Again girth 5 implies there is no 4-cycle. Thus if $z \neq x$ were adjacent to both y_i and y_j for $i \neq j$, then $x - y_i - z - y_j - x$ would be a 4-cycle, a contradiction. Therefore the sets

$$N(y_i) \setminus \{x\} \quad (i = 1, \dots, t)$$

are pairwise disjoint.

Step 3: counting. All vertices in

$$\{x\} \cup N(x) \cup \bigcup_{i=1}^t (N(y_i) \setminus \{x\})$$

are distinct, so

$$|V(G)| \geq 1 + t + \sum_{i=1}^t |N(y_i) \setminus \{x\}| = 1 + t + \sum_{i=1}^t (\deg(y_i) - 1).$$

Since $\deg(y_i) \geq d$ for all i , we get $\deg(y_i) - 1 \geq d - 1$, hence

$$|V(G)| \geq 1 + t + t(d - 1) = 1 + td \geq 1 + d^2.$$

This proves $|V(G)| \geq d^2 + 1$.

Step 4: the equality case. If $|V(G)| = d^2 + 1$, then all inequalities above are tight. In particular, $\deg(x) = t = d$ and $\deg(y_i) = d$ for all i . Repeating the argument with x replaced by an arbitrary vertex shows that G is d -regular. Tightness also forces that every vertex $z \notin \{x\} \cup N(x)$ is adjacent to *exactly one* neighbor of x (otherwise we would either miss vertices or create a 4-cycle). Equivalently, any two nonadjacent vertices have exactly one common neighbor, while adjacent vertices have none (otherwise a triangle would appear). Hence G is strongly regular with parameters

$$(v, k, \lambda, \mu) = (d^2 + 1, d, 0, 1).$$

Such graphs are precisely the Moore graphs of girth 5.

Step 5: why $d = 4$ cannot have equality. Assume for contradiction that equality holds for $d = 4$. Then G would be strongly regular with

$$(v, k, \lambda, \mu) = (17, 4, 0, 1).$$

For a strongly regular graph, the two nontrivial adjacency eigenvalues are

$$r, s = \frac{(\lambda - \mu) \pm \sqrt{(\lambda - \mu)^2 + 4(k - \mu)}}{2},$$

and their multiplicities are

$$m_r = \frac{1}{2} \left((v-1) - \frac{2k + (v-1)(\lambda - \mu)}{\sqrt{\Delta}} \right), \quad m_s = \frac{1}{2} \left((v-1) + \frac{2k + (v-1)(\lambda - \mu)}{\sqrt{\Delta}} \right),$$

where $\Delta = (\lambda - \mu)^2 + 4(k - \mu)$. Substituting $(v, k, \lambda, \mu) = (17, 4, 0, 1)$ gives $\Delta = 13$ and $2k + (v-1)(\lambda - \mu) = 8 + 16(-1) = -8$, hence

$$m_r = \frac{1}{2} \left(16 + \frac{8}{\sqrt{13}} \right), \quad m_s = \frac{1}{2} \left(16 - \frac{8}{\sqrt{13}} \right),$$

which are not integers. This contradicts that eigenvalue multiplicities must be integers. Therefore no such graph exists, and equality cannot hold when $d = 4$. $\square$

Remark 3.1.3. Known examples achieving $|V(G)| = d^2 + 1$ occur for $d = 2$ (the cycle C_5), $d = 3$ (the Petersen graph), and $d = 7$ (the Hoffman–Singleton graph). A hypothetical case $d = 57$ is a famous open problem; for all other d equality is impossible.

Exercise 3.1.4. Let $Q = \{1, \dots, q\}$ be the graph with vertex set Q^n , two vertices adjacent if they differ in exactly one coordinate. Show that Q is Hamiltonian.

证明. We prove the statement by induction on n .

Base case $n = 2$. The vertices are pairs $(a, b) \in Q^2$. Consider the cycle

$$(1, 1), (2, 1), \dots, (q, 1), (q, 2), (q-1, 2), \dots, (1, 2), (1, 3), (2, 3), \dots,$$

continuing in this “serpentine” fashion through the rows $b = 1, 2, \dots, q$, and finally ending at $(1, q)$ and returning to $(1, 1)$. Consecutive vertices differ in exactly one coordinate, and every vertex of Q^2 appears exactly once, so this is a Hamilton cycle in $H(2, q)$.

Inductive step. Assume $H(n-1, q)$ is Hamiltonian for some $n \geq 3$. Fix a Hamilton cycle in $H(n-1, q)$ and write it as

$$w_1, w_2, \dots, w_m, w_1, \quad \text{where } m = q^{n-1},$$

so that w_i, w_{i+1} differ in exactly one coordinate for each i (indices modulo m).

We now construct an explicit Hamilton cycle in $H(n, q)$. For each $i = 1, \dots, m$, define a path P_i inside the fiber $\{(a, w_i) : a \in Q\} \subseteq Q \times Q^{n-1}$ by

$$P_i := \begin{cases} (1, w_i), (2, w_i), \dots, (q-1, w_i), & \text{if } i \text{ is odd,} \\ (q-1, w_i), (q-2, w_i), \dots, (1, w_i), & \text{if } i \text{ is even.} \end{cases}$$

Thus P_i visits exactly the vertices (a, w_i) with $a \in \{1, \dots, q-1\}$, each exactly once, and consecutive vertices in P_i differ only in the first coordinate, hence are adjacent in $H(n, q)$.

Next, concatenate these paths in the order $P_1, P_2, \dots, P_m$. The last vertex of P_i and the first vertex of P_{i+1} have the same first coordinate (both are $(q-1, *)$ or both are $(1, *)$ by construction), and their Q^{n-1} -parts are w_i and w_{i+1} , which differ in exactly one coordinate. Hence the connecting edge between P_i and P_{i+1} is also an edge of $H(n, q)$ (it changes only one of the last $n-1$ coordinates).

So far we have produced a Hamilton *path* through all vertices (a, w) with $a \in \{1, \dots, q-1\}$ and $w \in Q^{n-1}$, i.e. through $(q-1)q^{n-1}$ vertices.

Finally, append the remaining q^{n-1} vertices with first coordinate q in the order

$$(q, w_m), (q, w_{m-1}), \dots, (q, w_1),$$

and then close the cycle by returning to $(1, w_1)$. This works because:

- The last vertex of P_m is either $(1, w_m)$ or $(q-1, w_m)$, and in either case it is adjacent to (q, w_m) since only the first coordinate changes.
- For each j , the vertices (q, w_j) and (q, w_{j-1}) are adjacent because w_j and w_{j-1} differ in exactly one coordinate in $H(n-1, q)$.
- The vertex (q, w_1) is adjacent to $(1, w_1)$ since only the first coordinate changes.

Thus we obtain a cycle in $H(n, q)$.

Moreover, every vertex of Q^n appears *exactly once* in this cycle: each (a, w_i) with $a \leq q-1$ appears exactly once in the concatenation of the P_i , and each (q, w_i) appears exactly once in the final appended segment. Hence the cycle is Hamiltonian.

This completes the induction and the proof. $\square$

Exercise 3.1.5. Find the six nonisomorphic trees on 6 vertices, and for each compute the number of distinct spanning trees in K_6 isomorphic to it.

- Exercise 3.1.6.** (a) Suppose a tree G has exactly one vertex of degree i for $2 \leq i \leq m$ and all other vertices have degree 1. How many vertices does G have?
- (b) Suppose a tree G has exactly one vertex of degree i for $2 \leq i \leq m$ and k other vertices have degree 1. Prove that $k \geq \lceil \frac{m+3}{2} \rceil$ and give a construction for such a graph.